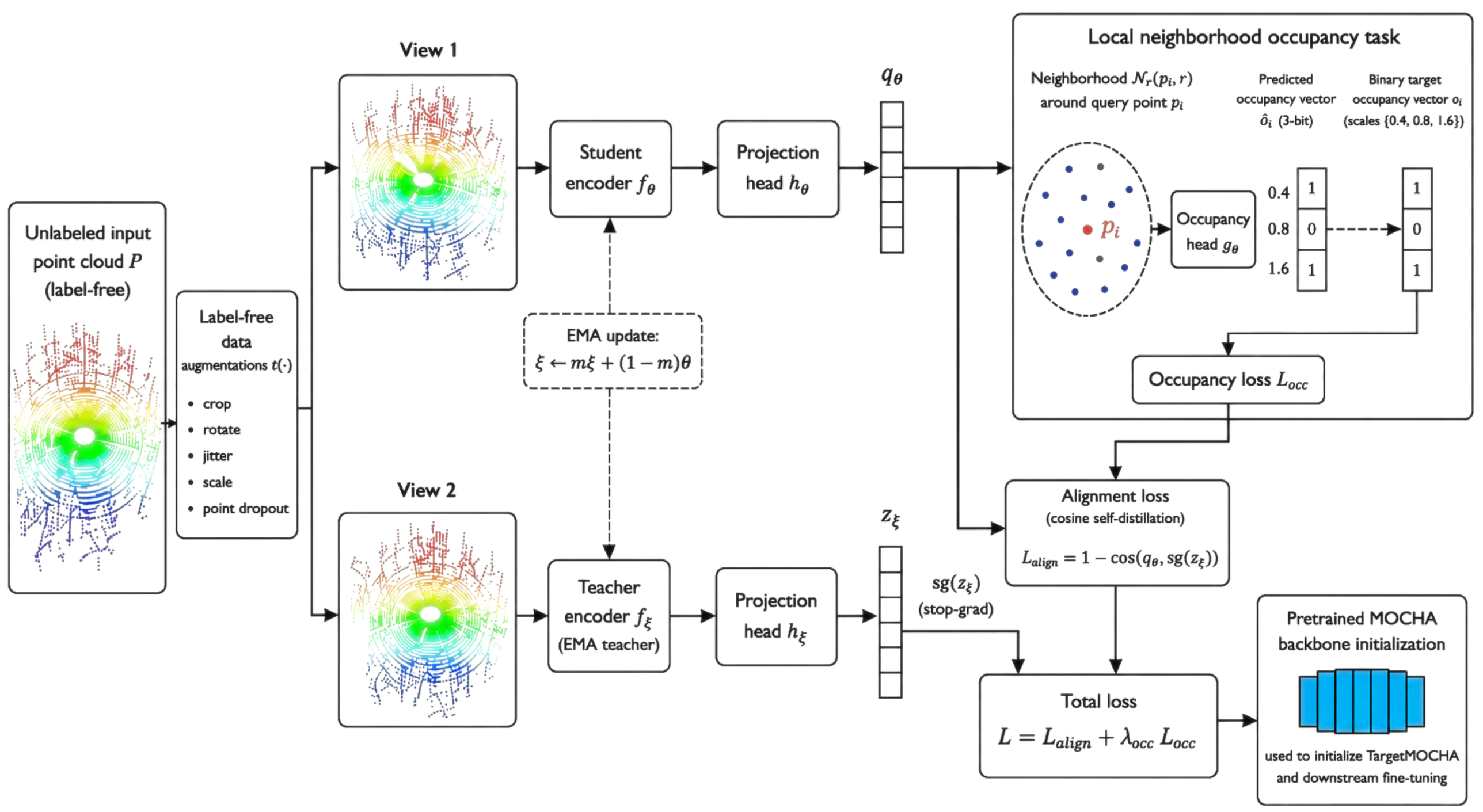

Unlabeled input point cloud P (label-free)
Label-free data augmentations $t(\cdot)$
crop
rotate
jitter
scale
point dropout
View 1
View 2
Student encoder f_θ
Teacher encoder f_ξ (EMA teacher)
EMA update: $\xi \leftarrow m\xi + (1-m)\theta$
Projection head h_θ
Projection head h_ξ
q_θ
z_ξ
$\mathrm{sg}(z_\xi)$ (stop-grad)
Local neighborhood occupancy task
Neighborhood $\mathcal{N}_r(p_i, r)$ around query point p_i
Predicted occupancy vector $\hat{o}_i$ (3-bit)
Binary target occupancy vector o_i (scales {0.4, 0.8, 1.6})
p_i
Occupancy head g_θ
0.4
0.8
1.6
1
0
1
1
0
1
Occupancy loss L_{occ}
Alignment loss (cosine self-distillation)
$L_{align} = 1 - \cos(q_\theta, \mathrm{sg}(z_\xi))$
Total loss
$L = L_{align} + \lambda_{occ} L_{occ}$
Pretrained MOCHA backbone initialization
used to initialize TargetMOCHA and downstream fine-tuning